# Calcurion Maths Worksheets

A-Level Mathematics

**Topic:** Statistics – Probability

Medium & Hard • 25 Questions

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## Section A – Medium (Q1–15)

**1.** A fair die is rolled. Find $P(\text{even})$. **[1]**

**2.** A card is drawn from a standard pack. Find $P(\text{Ace})$. **[2]**

**3.** $P(A) = 0.4$, $P(B) = 0.5$, $A$ and $B$ mutually exclusive. Find $P(A \cup B)$. **[2]**

**4.** $P(A) = 0.3$, $P(B) = 0.6$, $A$ and $B$ independent. Find $P(A \cap B)$. **[2]**

**5.** $P(A) = 0.5$, $P(B) = 0.4$, $P(A \cap B) = 0.2$. Find $P(A \cup B)$. **[2]**

**6.** $P(A) = 0.6$, $P(A \cap B) = 0.2$. Find $P(B|A)$. **[3]**

**7.** Two fair coins are tossed. Find the probability of two heads. **[2]**

**8.** A bag contains 5 red and 3 blue balls. One is drawn at random. Find $P(\text{red})$. **[1]**

**9.** From the same bag, two balls are drawn without replacement. Find $P(\text{both red})$. **[3]**

**10.** From the same bag, two balls are drawn with replacement. Find $P(\text{both red})$. **[2]**

**11.** $P(A) = 0.7$. Find $P(A')$. **[1]**

**12.** Events $A$ and $B$ are independent with $P(A) = 0.4$, $P(B) = 0.5$. Find $P(A \cup B)$. **[3]**

**13.** A spinner has sections 1, 2, 3, 4 with $P(1) = 0.1$, $P(2) = 0.2$, $P(3) = 0.3$. Find $P(4)$. **[2]**

**14.** Given $P(A|B) = 0.6$, $P(B) = 0.5$. Find $P(A \cap B)$. **[2]**

**15.** Draw a Venn diagram for two events with $P(A) = 0.4$, $P(B) = 0.5$, $P(A \cap B) = 0.15$. Find $P(A \cup B)$. **[3]**

## Section B – Hard (Q16–25)

**16.** A bag contains 4 red, 3 green and 5 blue balls. Two are drawn without replacement. Find $P(\text{same colour})$. **[5]**

**17.** $P(A) = 0.5$, $P(B) = 0.6$, $P(A \cup B) = 0.8$. Find $P(A \cap B)$ and determine whether $A$ and $B$ are independent. **[4]**

**18.** Three fair dice are rolled. Find the probability that all three show different numbers. **[4]**

**19.** A and B play a game. $P(A \text{ wins}) = 0.6$. They play until one wins two games in a row. Find $P(A \text{ wins the match})$. **[6]**

**20.** Use a tree diagram: first bag 3R 2B; second bag 4R 1B. A bag is chosen at random, then a ball. Find $P(\text{red})$. **[5]**

**21.** Given the above, find $P(\text{bag 1}|\text{red})$. **[4]**

**22.** $P(A) = 0.4$, $P(B|A) = 0.3$, $P(B|A') = 0.5$. Find $P(B)$. **[4]**

**23.** Prove that if $A$ and $B$ are independent then $A$ and $B'$ are independent. **[4]**

**24.** A family has two children. Given that at least one is a boy, find the probability that both are boys (assume equal probability). **[4]**

**25.** Events $A$, $B$, $C$ are mutually exclusive and exhaustive. $P(A) = 2P(B) = 3P(C)$. Find $P(A)$. **[4]**

*End of Worksheet* • *Show all working clearly.*